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Raihanah Adawiyah Shaharuddin and Md Yushalify Misro

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Traffic Simulation Using Agent Based Modelling

Raihanah Adawiyah Shaharuddin[a)] and Md Yushalify Misro[b)]

School of Mathematical Sciences, Universiti Sains Malaysia, 11800 USM Penang, Malaysia

[a)]*Corresponding author: raihanahadwyh@gmail.com*
[b)]*yushalify@usm.my*

Abstract. This paper studies two different models of agent-based modelling in simulating an urbanized city transportation network under various conditions and situations. Urbanized city traffic requires consideration of wide mobility modes to better represents the impact of certain real-life situations. Agent-based modelling with powerful modelling and simulation platforms offers an amazing framework to fulfil the needs of modelers to study from nanoscopic to macroscopic phenomena. Over the years, many platforms have been developed to cope with the need of simulating more complicated agent-based model. With GAMA (GIS & Agent-based Modelling Architecture), an open-source (GPL) tool that allows modelling of the real-life situation with a simpler framework, simulating the urban traffic involving the area around Universiti Sains Malaysia (USM).

INTRODUCTION

Urban cities are facing extreme congestion with increasing vehicles over the years. To handle such phenomena, many cities have opted to diversify their urban travels by implementing green transport (walking, bicycles), surface public transport (buses, tramways), and rapid transit (metro). Implementing it into reality requires a clear transport policy that involves high cost and heavy public works. For policymakers or even urban planners, decision support tools help them to investigate existing problems and determine the appropriate solutions. Simulation tools are a better option than optimization tools to elaborate on a multimodal transport policy while considering the final users profile [1, 2].

Recently, the study about the maximum speed limit of highway using different type of curves [3, 4, 5], researches on highway designing using spiral curves [6, 7], variety of transition curves [8, 9, 10] and computational degree of road elevation [11] not enough to convince the policy maker because the traffic problem itself are complex and need special procurement methods that can help them to make effective and accurate decision. This is because the traffic problem comprise so many factors and variables.

For tools like simulation, representation, and interpretation, the use of Agent-Based Models (ABM) has increased in demand in constructing a more complex models. In [12], a complex system is defined as a dynamic system with multiple interaction agents whose emergent behaviour cannot be simply inferred from the behaviour of the agents. In the system, the combined interactions generate into large-scale behaviour and unable to be estimated simply by knowing the behaviour of any single agent. Urban city transportation network is considered as a complex system as it comprises of multiple agents (people, vehicles, buildings, road networks, etc) so ABM is a great option.

ABM is a computational modelling paradigm that allows us to simulate the individual actions of multiple agents and calculate the resulting system behaviour over time and many platforms have been developed for a more complex ABM (MATSim, Sumo, SimMobility, Netlogo, etc). Netlogo has a modelling language feature that eases the model design but the shortcoming is not well adapted to handle a large-scale traffic model that relies on GIS data. Meanwhile, the Repast platform provides the user to design a rich and complex model but since it requires high computer science skills, many modellers depend on computer scientists to develop their agent-based models. Therefore, in 2007, GAMA (GIS & Agent-based Modelling Architecture) was introduced as an alternative platform to overcome these limitations.

GAMA enables the use of complex GIS data that will serve as an environment for the agents. GAMA platform 1.6 version is used for this study and the features available are explained through [13, 14]. [15] has introduced a new plug-in feature that is available in the GAMA platform which allows the user to simulate by incorporating the road infrastructures and enable the vehicle to change the lane of preference.

USM main campus is located in Penang, Malaysia where it is among the highest population density and one of the nation's urbanized states. Urban traffic congestion will influence the traffic inside the USM main campus. In 2018 alone, there are almost 4000 USM students that are graduating and a mass crowd is expected throughout the convocation week. Hence, effective measures are needed to improve the mobility mode and transportation network on the campus. The sudden spike of traffic demand must be overcome with an effective measure such as by transforming more efficient road networking during graduation week. Therefore, this paper studies the efficiencies of the traffic flow by manipulating the path of the road network in the USM main campus. With the integration of GIS data, GAMA provides distinguish agents environment for two different models. These models, USM Normal Traffic and USM Graduation Traffic are based on a framework of the model in [15].

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PRELIMINARIES AND PLUG-INS

Extraction of Data

GAML operates completely by using the meta-model where it consists of three main sets of abstract classes which are space, entities, and time. The space of a model is mainly composed of two main classes which are geometry that shapes the embodiment of an agent in its environment and topology that is forming the environment itself. The study is conducted by identifying the area within and around the USM main campus as the environment of interest. This area network topology is examined for a better representation of the traffic scenario in real life. Network topology is the connection of the elements that form a communication network which in this study the nodes (*intersection*), links (*road*), shape geometry (*building*) are the elements needed. The elements embedded in the network file is downloaded through Open Street Map (OSM). From the OSM file, the attributes (description of a defined agent and its behaviours) are then manipulated by using a tool named QGIS (Quantum Geographic Information System) to meet the expectation of the traffic scenario and the purpose of the study. Then, these elements are exported in the format of shapefile to be included in the model of the GAMA project.

Construction of Environment

Each road is a polyline composed of road sections (*segments*). At this step, each link that represents the road network (*roads shapefile*) ensures that the properties such as the maximum speed (*maxspeed*), number of lanes (*lane*), and the direction of the lane (*oneway*) are accurate with the actual traffic networks and rules.
•*oneway*: The direction of the lane is either bidirectional or one-way.
•*lane*: The number of lanes follows the actual lane of real traffic. The number of lanes varies from 1 to 4 lanes.
•*maxspeed*: The *maxspeed* is the maximum speed of every lane. The maximum speed range from 30 km/h to 50 km/h and is based on the number of lanes. The maximum speed of 1 lane road is 30 km/h, 2 lane road is 40 km/h and lane of the road above 2 is 50 km/h.

The nodes networks are a collection of the road networks vertex where each segment will start with a source node (*source_node*) and ends with a target node (*target_node*), hence the input and output of the road could be defined. Other than that, it also defines the nodes for intersections. For the USM Graduation Traffic, it is based on the GIS road data of USM Normal Traffic but some parts will be transformed by altering the direction of the road (bidirectional to one way). Consequently, it changes the way the graph of the road links with each other. Every segment of the shapefile becomes an agent where in this case the road shapefile will be presenting the road agents for the models. These segments of the road agent will act as a driving graph that is connected by vertices (*nodes shapefile*) where the moving entities will use it as a path.

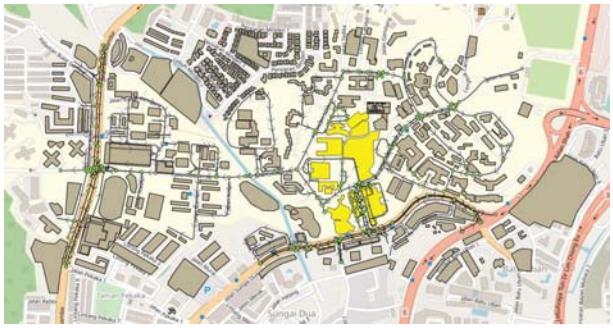


FIGURE 1. The compilation of roads, nodes and building shapefiles in constructing the environment of the models.

Figure 1 shows the combinations of the *shapefiles* which creates an environment for our project simulations. The building is related to human activities throughout everyday life wherein this study, the buildings mainly involving the build-up structures in USM main campus and the area around it. Buildings are generated from the shapefile (*buildings.shp*) in the GAMA platform to create the building agents. The building agent is important for the initialization location of the moving entities. At initialization, the people agent (*moving entities*) are located at the building agent

that is defined as "DTSP"(yellow polygons). The space of the GAMA meta-model involved the *building, roads*, and *nodes shapefiles* to create an ideal traffic scenario that matches with the purpose of this project.

ALGORITHM AND SIMULATION

Creation of Entities

A meta-model for GAML needs entities as part of the basic set to develop the ABM. The agent is one of the representations of entities as it represents an individual entity of a simulation that is conceptually similar to an object in an object-oriented language [13]. Meanwhile, a group of agents that shares the same structure and behaviour is classified as a population. In this project model, a species of agents called people is defined. Each agent must be provided with an agenda, before deciding their path and move to the respective destination according to its behaviour. To execute such movement, the skill for the people agent is assigned where in this project *advanced driving skill* is used.

The *advanced driving skill* allows the vehicle properties and the personality of the agents to be defined. [13] proposed a more detailed explanation regarding the additional plugins that is built inside the *advanced driving skill*. In real life situations, the people agent is presenting the vehicle that moves according to the weight on the road network which contains the individual behaviour of the driver.

The number of people agent is created depending on the value of *nb_people*. At initialization, these agents are placed in one of the buildings that have the attribute of type "DTSP". Each vehicle has a length of 3m; they are driving on the right side, their probability of changing lane for an upper lane is randomly between 0.1 and 1; their probability of changing lane for a lower lane is randomly between 0.5 and 1; the security distance coefficient is randomly between 1 and 3; their probability to respect priorities is randomly between 0.8 and 1; their probability to respect stop signal is 1; their probability to block a node is 0; the maximal acceleration of their vehicle is 1.39; the speed coefficient of the driver is randomly between 0.8 to 1.2.

When the agent has no final target, the reflex *time to go* is activated. In this reflex, the agent will randomly choose one of the intersection nodes as its final target and computed the path to reach its target by following the traffic rules of the *road network* graph. Note that, in this process, each agent knows about the properties of the road network that they will be using (*the maximum speed, the number of lanes, the direction of the pathway, etc.*). Since the agent already sets a final target and has computed the path, then it will proceed to the *move* reflex which is followed by *do drive* action of driving the agent toward the direction of its final target. The people agent also able to decide whether it will be using the linked road when there is traffic congestion. This action is executed if the speed of the vehicle falls below 5 km/h.

Plot the Data for Average Speed (km/h) Against Number of Cycles

We are interested in the average speed (km/h) as in Equation (1) for all moving agents over time. Hence, the *mean* function is used to find the average speed and the value obtained is in terms of m/s. Then the value will be converted into km/h by multiplying it by 3.6.

$$\text{Average speeed } (km/h) = \frac{\text{Sum of speed for all people}}{\text{Number of people}} \tag{1}$$

Agent-Based Traffic Simulation Models Using GAMA

To simulate the traffic for people agent, the required steps are as follows:
Step 1: Loading the GIS data into GAMA project to build the environment.
Step 2: Place the initial location for agents and their targeted destination.
Step 3: Set the parameters for the number of moving agents (*nb_people*).
Step 4: Running the Agent-based simulation using GAMA.
Step 5: Counting the number of cycles (time) for all moving agents to be in each state (A, B and C).
Step 6: The simulation is repeated for multiple times to find the average number of cycles as in step 5.



(a) A: Initial **(b)** B: Intermediate **(c)** C: Final

FIGURE 2. The three states (A, B, C) of the people agent movement in the USM Normal Traffic simulation.



(a) A: Initial **(b)** B: Intermediate **(c)** C: Final

FIGURE 3. The three states (A, B, C) of the people agent movement in the USM Graduation Traffic simulation.

Step 7: Repeat steps 3 to 5 for different values of *nb_people*.
Step 8: Tabulate the average number of cycles for different values of *nb_people*.

The simulation results of GAMA are the average time taken for the people to move from the initial location to be in each state. The classification of conditions for the agent states is as shown in Fig. 2. Figure 2 (A) is when the people at the initial location, cycle=0, (B) when all the people have entered the main road while (C) is when none of the people moving on the road of USM main campus. For this scenario, we run two simulations for different *nb_people*=200 and 400. Each simulation is conducted five times and the average cycle needed for the moving agents to undergo transitions from the state (A) to (C) is recorded.

In USM Graduation Traffic, the path for the road network has been changed by referring to the path to USM main campus during graduation week. Hence, by altering the road graph, we will see how road networking will influence the road traffic in USM main campus by simulating the model using the Agent-based simulation tool, GAMA. In the USM Graduation Traffic simulation, it follows the same steps as in the earlier scenarios and they were classified into distinct states such as in Fig. 3. Two simulations will also be conducted for this scenario with the parameters *nb_people*=200 and *nb_people*=400. From there, the behaviour of the agents is observed and tabulated to be analyzed.

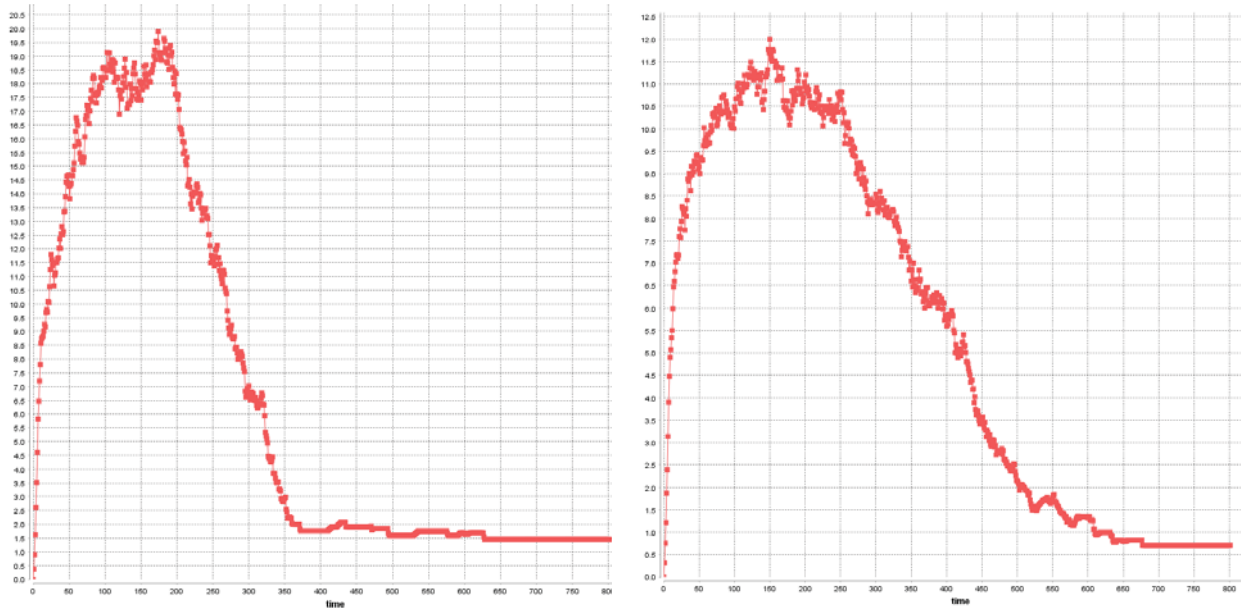
RESULT AND DISCUSSION

USM Normal Traffic model is simulated with a different value of *nb_people*=200 and 400 with each simulation step (cycle) is one second. Table 1 shows USM Normal Traffic with *nb_people*=200, at initialization the average time taken for all the agents to move to state B is 172 simulation cycles and 333 cycles from state A to C. Hence, it takes an average of 172 cycles for all agent to enter the main road and 333 cycles for them leave the USM main campus. Meanwhile, for the scenario with *nb_people*=400, from initialization state A to B it took an average of 443 cycles and 595 cycles from state A to C. Hence, it takes an average of 443 cycles for all agent to enter the main road and 595 cycles for them leave the USM main campus. If we were to compare the transition from A to C for both *nb_people*, it can be said that if we double the number of people from 200 to 400, approximately the average time taken from A to B magnified by 2.5 times and the average time taken from A to C magnified by 2 times.

Figure 4 (a) have an average speed of 18 km/h and reaches its peak at 20 km/h around 150 cycles, while Fig. 4 (b) reached an average speed of more than 11 km/h and reach its peak at 12 km/h when it is at 150 cycles. After 200 simulation steps, the average speed in both graphs began to decline gradually as the moving agents are experiencing

TABLE 1. Average of cycles for each different state for USM Normal Traffic simulation model.

nb_people	**200**			**400**		
State / **No. of repetition**	A	B	C	A	B	C
1	0	160	316	0	440	598
2	0	152	352	0	455	616
3	0	180	309	0	470	617
4	0	154	348	0	436	587
5	0	213	337	0	414	554
Average cycle	**0**	**171.8**	**332.4**	**0**	**443**	**594.4**



(a) *nb_people*=200 **(b)** *nb_people*=400

FIGURE 4. The graph of the average speed (km/h) against the number of cycles for USM Normal Traffic at different values of *nb_people*.

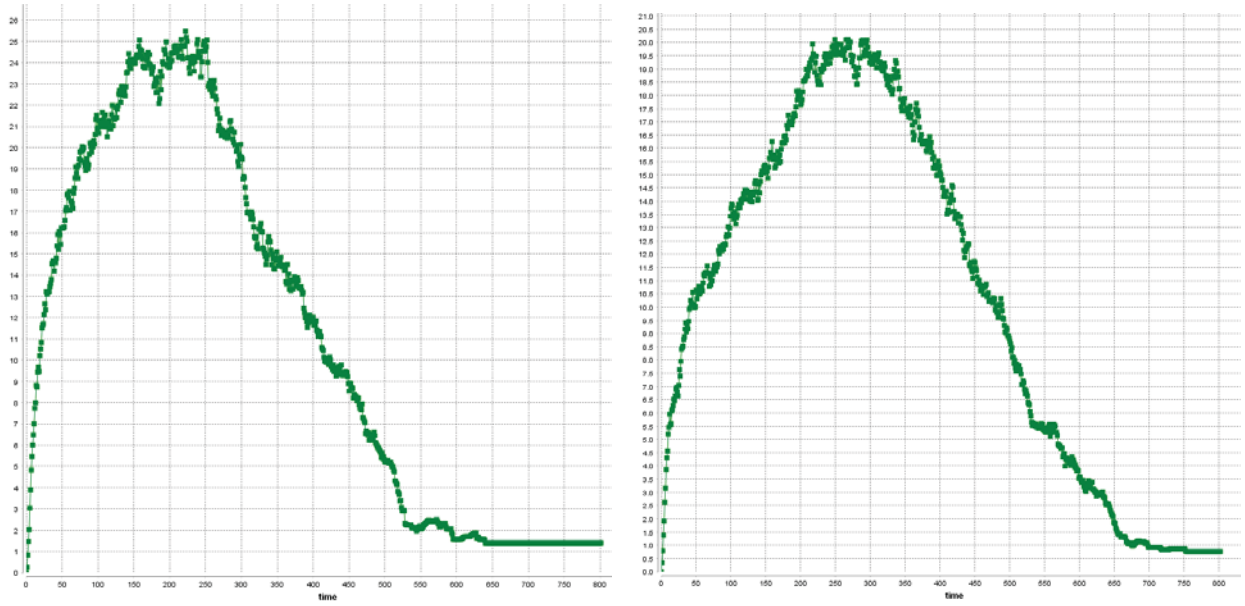
congestion and some of them have arrived at their respective final target (destination). Towards the end, as more and more agents have reached their destination and stop moving, the graph average speed began to reach 0 km/h. As we compare Fig. 4 (a) have a higher maximum average speed than Fig. 4 (b). Plus, during 100 to 200 simulation steps, Fig. 4 (a) already reach an average speed of more than 18 km/h which is higher than Fig. 4 (b). Therefore, by increasing the number of moving agents, the average speed with respect to time also decreases for USM Normal Traffic simulation.

In the USM Graduation Traffic simulation model, the number of cycles for all the people agents to be in states A, B and C are noted and the average cycle is calculated. Table 2 shows the comparison of the average cycle between the two values of *nb_people* for this model while Fig. 5 shows the graph of the average speed (km/h) against the number of cycles for USM Graduation Traffic.

In Table 2, for *nb_people*=200, it requires an average of 148 simulation cycles for the model to move from state A to B and 553 cycles for the model to move from state A to C. Next, simulation with *nb_people*=400, the moving agents are in state B after an average of 275 simulation steps and all the agents later in state C after an average of 744 simulation steps. Therefore, the average cycle needed for all 400 people to exit from the building and enter the main road is around 2 times higher compared to 200 people. Whereas, the average number of cycles required by the 400 people to leave the USM main campus road network is around 1.5 times higher than the average number of cycles

TABLE 2. Average of cycles for each different state for USM Graduation Traffic simulation model.

nb_people	**200**			**400**		
State / **No. of repitition**	A	B	C	A	B	C
1	0	154	592	0	283	813
2	0	142	539	0	288	734
3	0	149	566	0	270	738
4	0	142	572	0	249	718
5	0	150	494	0	315	717
Average cycle	**0**	**147.4**	**552.6**	**0**	**275**	**747**

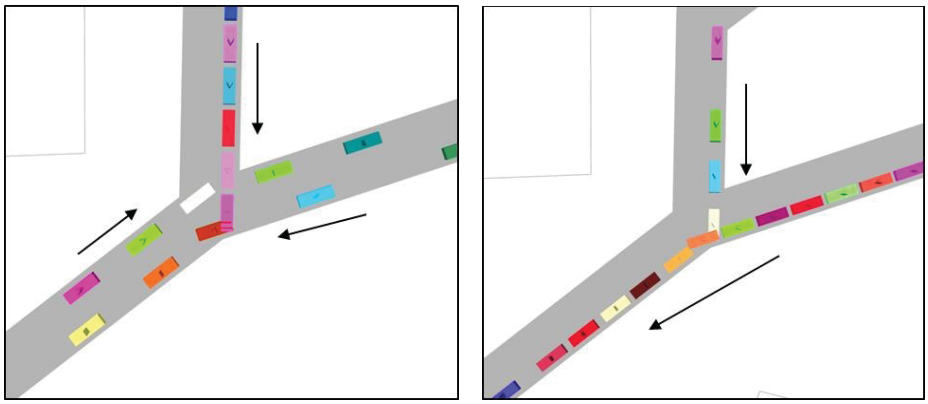


(a) *nb_people*=200 **(b)** *nb_people*=400

FIGURE 5. The graph of the average speed (km/h) against the number of cycles for USM Graduation Traffic at different values of *nb_people*.

needed by 200 people.

From the graphs in Fig. 5, when the simulation starts, the graph increases steadily as the vehicle began to move to their destinations on the road network. In Fig. 5 (a) the average speed increases over 20 km/h when the simulation is between 100 to 300 cycles and top the average speed of over 25 km/h as the simulation is around 200 to 250 cycles. Meanwhile, Fig. 5 (b) hit an average speed of 20 km/h when the simulation is running from 200 to 300 cycles and comes to a climax with an average speed higher than 20 km/h but less than 25 km/h. Then, the average speed drops steadily after 300 simulation steps because the moving agents are in a traffic jam that slows down the speed and some of them have arrived at their respective final target (destination). Ultimately, as most of the agents have arrived at their destination and stop moving, the graph average speed began to touch 0 km/h. By comparing both graphs in Fig. 5, we can see that (a) have 100 cycles more than (b) when the average speed is over 20 km/h. Plus, (a) has a maximum average speed at around 25 km/h higher than (b) with a maximum average speed of around 20 km/h. Hence, it can be observed that as the number of moving agents increase, the average speed decreases over time for USM Graduation Traffic simulation.



(a) USM Normal Traffic **(b)** USM Graduation Traffic

FIGURE 6. (a) A snapshot of a junction in USM Normal Traffic 3D simulation where moving agents are in the process to get onto the bidirectional main road. (b) A snapshot of a junction in USM Graduation Traffic 3D simulation where moving agents are in the process to get onto the one-way main road. The black arrow is indicating the direction of the road.

To model Agent-based simulation for the road traffic in USM main campus, agents have many behaviours depending on many factors. The movement of the agent is influenced by the surrounding infrastructure (road network), the position of other moving agents, the decision of crossing the road, accelerate to their desired speed, and many more. Figure 6 (a) and (b) show the snapshot of a junction of the two scenarios in the 3-D simulation where the transition from A to B occurs.

Figure 6 (a) shows the movement of the agent crossing a junction into the bidirectional main road. For this situation, each agent will first check if there is at least one vehicle coming from the right road at a distance lower than its security distance coefficient (1m to 3m). If there is none, it tests its probability to respect priorities (1.0: agent always respect priorities). Since it is a bidirectional road, so they need to consider vehicles that are coming from the left side at the same time. If there is no vehicle coming from the left and right side that is lower than its security distance coefficient, then the agent can exit the junction and enter the main road.

For the case in Fig. 6 (b), the agent only considers the vehicles coming from the left side of the road since it is a one-way road before crossing the intersection. Therefore, this explains why the simulation for USM Normal Traffic required more simulation steps for all the agents to enter the main road compared to the USM Graduation Traffic simulation. However, the average cycle for the people agent to undergo a transition from state A to state C is longer for USM Graduation Traffic compared to USM Normal Traffic. This implies that the time taken for all the people agent in USM Graduation to leave USM main campus is higher compared to USM Normal Traffic. This is mainly because the manipulation of the road network causes the path travelled to leave the USM main campus by the agent is longer in USM Graduation Traffic. Although the manipulation of the road network from bidirectional to the one-way road makes the time taken for the moving agent to enter the main road shorter, this alteration causes the length of the travel path increases since the route is only in one direction.

Furthermore, it is observed that the maximum average speed for USM Normal Traffic is lower than USM Graduation for both values of *nb_people*. Congestion is one of the main factors that cause such situations to happen. Meanwhile, for USM Graduation Traffic it has a higher maximum average speed compared to the other model since the road is less congested. Less congestion means a better flow in the traffic. Hence, we can say that USM Graduation Traffic has a better traffic flow than USM Normal Traffic which means the one-way road is preferable when there are high usage and demand for a particular road network to ensure smooth traffic flow.

CONCLUSION

Changes in the route of the road network and density of user on the road impacted the behaviour of the moving entities. The resulted behaviour depends on many parameters and the parameter that is manipulated in the simulation is the number of moving agents (*nb_ people*). Thus, two simulations for both ABM models with a different number of moving agents were conducted where the outcome over time and the resulted behaviour were analyzed. We do the

simulations for each road network where we test for USM Normal Traffic and USM Graduation Traffic. The time of simulation for the one-way road network to cross the junction and get onto the main road is much shorter than that of the bidirectional road network. Hence, in the case where there are a lot of people who need to leave a building, the time taken for them to leave the building will be shorter and faster if the road is altered to a one-way road network rather than a bidirectional road network. However, this manipulation results in a longer path to reach their destination since the road will be in one direction only. Therefore, the time to leave the building is shorter in a one-way road network but the shortcoming is that they require a lot more time to reach their final destination. However, the one-way road has better traffic flow compared to a bidirectional road.

The outcome of this simulation will have better accuracy if it can be modelized to be as realistic as a real-life scenario. ABM is very expensive in computation so they require huge time running simulation as it considers every behaviour of the agent. To simplify the model and reduce the computational time, this project only considers one species of moving agent that is initially located in the USM campus, then head towards their respective final target. Therefore, for a more realistic simulation, users can consider adding other modes of transportation such as motorcycles, bus, and pedestrians. Moreover, the visual of the traffic simulation not only available in 2 dimensional but also can be visualized in the 3 dimensional. The movement of the people agent especially at the junction, the behaviour of the car changing their respective lanes and many more can be observed more clearly with 3D simulation. The disadvantage of the simulation represented in 3D is that it is slower than that in 2D since the computation time for building the 3D environment and species is longer. Hence, the visualization of this road traffic simulation is helpful for route planners to be as one of the decision tools.

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